

PREM ARVINTH G

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PROFESSIONAL SUMMARY

ECE graduate with project experience in IoT-based hardware systems and computer vision. Built a real-world smart streetlight fault detection system using ESP32 and sensors, and presented it at an international conference. Familiar with Python, circuit-level prototyping, and sensor integration. Looking to grow in an entry-level ECE or related role.

TECHNICAL SKILLS

Embedded & IoT	ESP32, Sensor Integration (CT, LDR)
Languages	Python, SQL
Software	OpenCV, YOLOv8, Tesseract OCR, Pandas, NumPy
Simulation	Wokwi
Tools	Git, VS Code, PythonAnywhere

PROJECTS

Automatic Smart Streetlight Fault Detection & Response System

2025 – 2026

ESP32, CT Sensor, LDR, Python, IoT, Wokwi

- Built an IoT-based system using ESP32 with CT and LDR sensors to monitor real-time current and light intensity data from streetlights.
- Simulated the circuit using Wokwi before physical prototyping; assembled and tested the hardware setup.
- Sensor data was used to automatically identify faulty streetlights, removing the need for manual inspection.
- Presented at an International Conference held at Peri College.

Automatic Number Plate Recognition — Computer Vision

2023

Python, YOLOv8, OpenCV, SORT, Tesseract OCR

- Built a Python pipeline to detect vehicles in video streams using YOLOv8, track them with SORT across frames, and extract license plate text via Tesseract OCR.
- Designed for real-world integration into traffic management and smart city infrastructure.

EDUCATION

B.E. Electronics & Communication Engineering — Anna University

2022 – 2026

Sri Venkateswaraa College of Technology | CGPA: 7.51 / 10

Higher Secondary (HSC) — Tamil Nadu State Board

2022

V.H.N Hr. Sec. School | 75.16%

SSLC — Tamil Nadu State Board

2020

V.H.N Hr. Sec. School | 66.66%

CERTIFICATIONS & ACHIEVEMENTS

- International Conference — Presented at Peri College (Smart Streetlight Fault Detection)
- Winner — Internal College Hackathon (Image Classification)
- IoT Workshop — IIT Madras (Mini Robotic Arm)
- Python for Everybody — University of Michigan, Coursera